



Fabio Slika Stella

Embedded Software Engineer
Embedded Linux & IoT
C++/C Developer

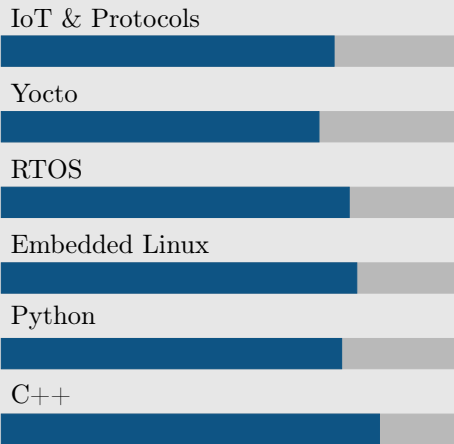
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About Me

Computer Engineer with hands-on experience in low-level software, Embedded Linux and real-time C++/C firmware. Career across access control, railway, automotive and industrial automation. Comfortable with Linux kernel and device-driver concepts, Yocto builds, RTOS and IoT communication protocols, with a consistent focus on reliability, automation and debugging close to the hardware.

Currently working on Embedded Linux and firmware for IoT access-control devices at ControlID – Yocto, bootloaders, OTA and C++ across TI and ST MCUs. Open Source side projects under stellacorp.com.br.

Skills



(*)[Scale 0 (Basic) a 6 (Advanced)]

Career Objective

Senior-level role in embedded software and IoT development, applying modern C++ and Python, Embedded Linux and real-time systems to architect reliable, maintainable connected devices.

Education

2014–2022 Bachelor's Degree in Computer Engineering Pato Branco
Federal University of Technology – Paraná (UTFPR)

Languages

Portuguese Native
English Advanced

Professional Experience

- 2026–Present Embedded Software Developer II ControlID
Embedded Linux and firmware for IoT access-control devices: Yocto builds, bootloaders and OTA pipelines, C++/C development across TI and ST MCUs, communication protocols, diagnostics and debugging.
- 2024–2025 Embedded Systems Engineer Progress Rail – Caterpillar
Low-level C/C++ firmware for safety-critical railway modules; system validation under operational and stress conditions; real-time constraints and industrial communication protocols.
- 2024 Software Engineer (Embedded) Zeentech / Stellantis
Embedded C/C++ modules for automotive systems; support in integration and validation; low-level timing and stability investigations.
- 2022–2024 Embedded Systems Developer InoBram
STM32 firmware and custom bootloader maintenance; UART, I2C and RS-485 protocol implementation; refactoring for stability and testability; in-circuit debugging.
- 2022 Test Engineer IAV Brazil
Automotive embedded validation with Typhoon HIL and black-box testing; failure documentation and root-cause analysis support.
- 2021–2022 R&D Intern Xpert Automação
Embedded firmware (ESP32) for fuel station automation; Linux-based systems, networking and OpenWRT integration.
- 2019–2022 Researcher UTFPR – Pato Branco
Modeling and simulation of control algorithms for electric motors; performance analysis and validation tools.

Technical Qualifications

- Modern C++ (C++17/20) with object-oriented design; C and Python for tooling and automation.
- Embedded Linux: Yocto builds, system bring-up, OTA pipelines and bootloaders.
- RTOS and real-time firmware (FreeRTOS, bare-metal) on STM32, ESP32, TI and ST MCUs.
- IoT and field protocols: MQTT, HTTP/IP, CAN/J1939, RS-485, I²C, SPI and UART.
- Linux-friendly tooling: Git, CMake, Docker, CI/CD and Shell/Python automation.
- Cross-platform / cross-development mindset with diagnostic rigor (GDB, JTAG/SWD, serial tracing).
- Kernel-adjacent foundation: memory, timing, interrupts, device-driver and boot flow concepts.
- Hands-on hardware bring-up and firmware–hardware integration with Ki-Cad.

Additional Courses

- 2023 Embedded Systems Bare-Metal Programming (STM32) Udemy
Online
- 2023 IoT Application Development with ESP32 (ESP-IDF) Udemy
Online
- 2020 Networking Academy Learn-A-Thon Cisco
Online
- 2020 Introduction to Cybersecurity Cisco
Online